Question	Answer	Marks	Partial Marks
6(b)(ii)	3136	**1**	
6(b)(iii)	47	**1**	
6(b)(iv)	1	**1**	
6(c)	156	**2**	**M1** for (12 =) 2, 2, 3 and (78 =) 2, 3, 13 or for 2, 2, 3, 13 or for list of multiples of 12 and 78 to at least 156 If 0 scored **SC1** for 156*k*
6(d)	6	**2**	**B1** for 2 or 3 as final answer
6(e)	$2^4 \times 3^3$ or $2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$	**2**	**M1** for a complete factor tree with at most one error or 2, 2, 2, 2, 3, 3, 3 clearly identified as factors or **B1** for a correct product that equals 432
7(a)	−1.2, −1.5, 1.5	**2**	**B1** for 2 correct
7(b)	Correct curve	**4**	**B3FT** for 11 or 12 points correctly plotted or **B2FT** for 9 or 10 points correctly plotted or **B1FT** for 6, 7 or 8 points correctly plotted
7(c)	1.2 to 1.4	**1**	**FT** *their* **(b)**
7(d)(i)	Correct ruled line	**1**	
7(d)(ii)	(−2.6 to −2.3, −2.6 to −2.3,) and (2.3 to 2.6, 2.3 to 2.6)	**2**	**FT** $y = x$ drawn and *their* curve **B1FT** for one correct, or both *x* values correct or both *y* values correct
8(a)(i)	$\frac{4}{20}$ oe	**1**	
8(a)(ii)	0	**1**	
8(a)(iii)	$\frac{17}{20}$ oe	**1**	
8(b)(i)	9.75 or 9.746 to 9.747	**3**	**M2** for $\sqrt{12^2 - 7^2}$ soi or **M1** for $7^2 + BC^2 = 12^2$ soi
8(b)(ii)	33.5 or 33.49 or 33.492 to 33.494	**1**	**B1 FT** *their* **(b)(i)** × 2 + 14

Question	Answer	Marks	Partial Marks
8(c)	Correct construction and region shaded	**5**	**B2** for arc radius 7.5 cm, centre *P* or **B1** for short arc radius 7.5 cm, centre *P* or **B1** for arc, any radius centre *P* **B2** for angle bisector with correct arcs or **B1** for angle bisector with incorrect or no arcs **B1** for region **dep** on **B1**(arc) and **B1**(bisector)
9(a)	Tangent	**1**	
9(b)	Radius drawn on circle	**1**	
9(c)(i)	90 radius [and] tangent [at 90]	**2**	**B1** for each
9(c)(ii)	90 angle [in a] semicircle [= 90]	**2**	**B1** for each
9(d)(i)	63.6 or 63.61 to 63.63	**2**	**M1** for $4.5^2 \times \pi$
	cm^2	**1**	
9(d)(ii)	5.16 or 5.162 …	**2**	**M1** for $\sin 35 = \frac{BC}{9}$ or better